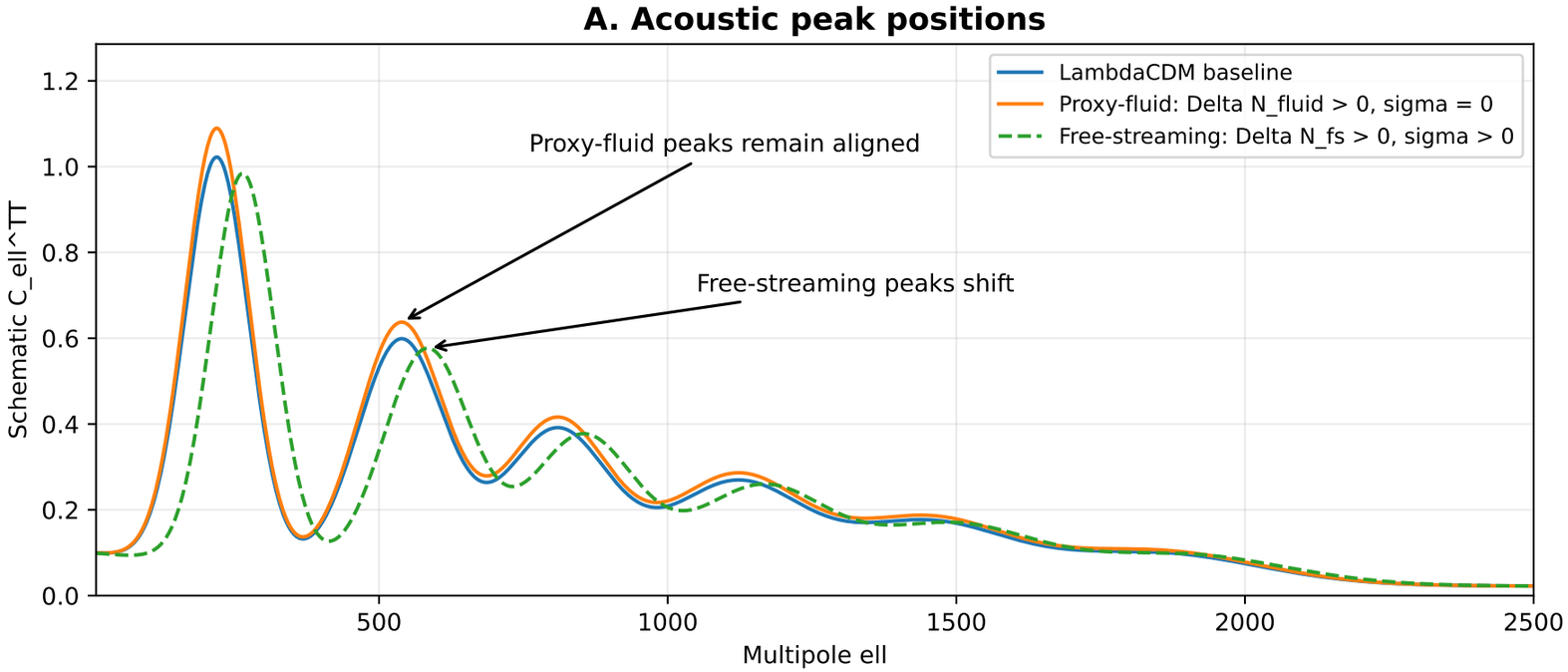
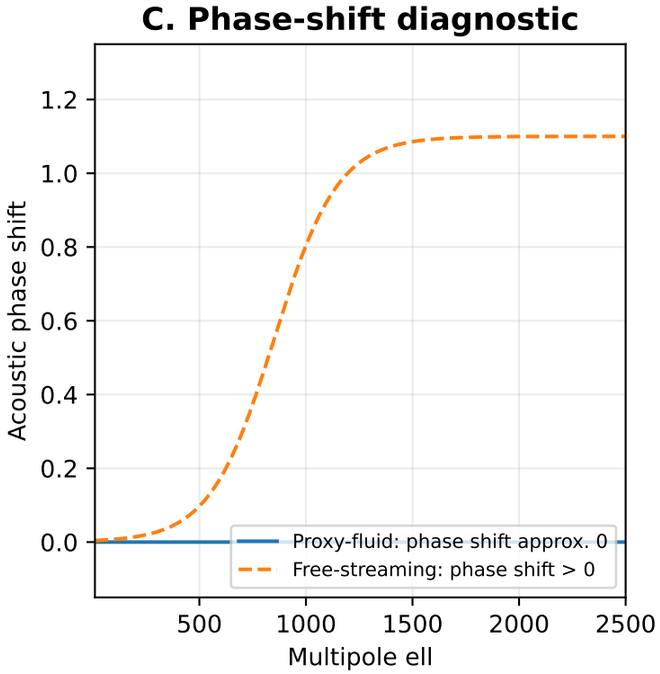
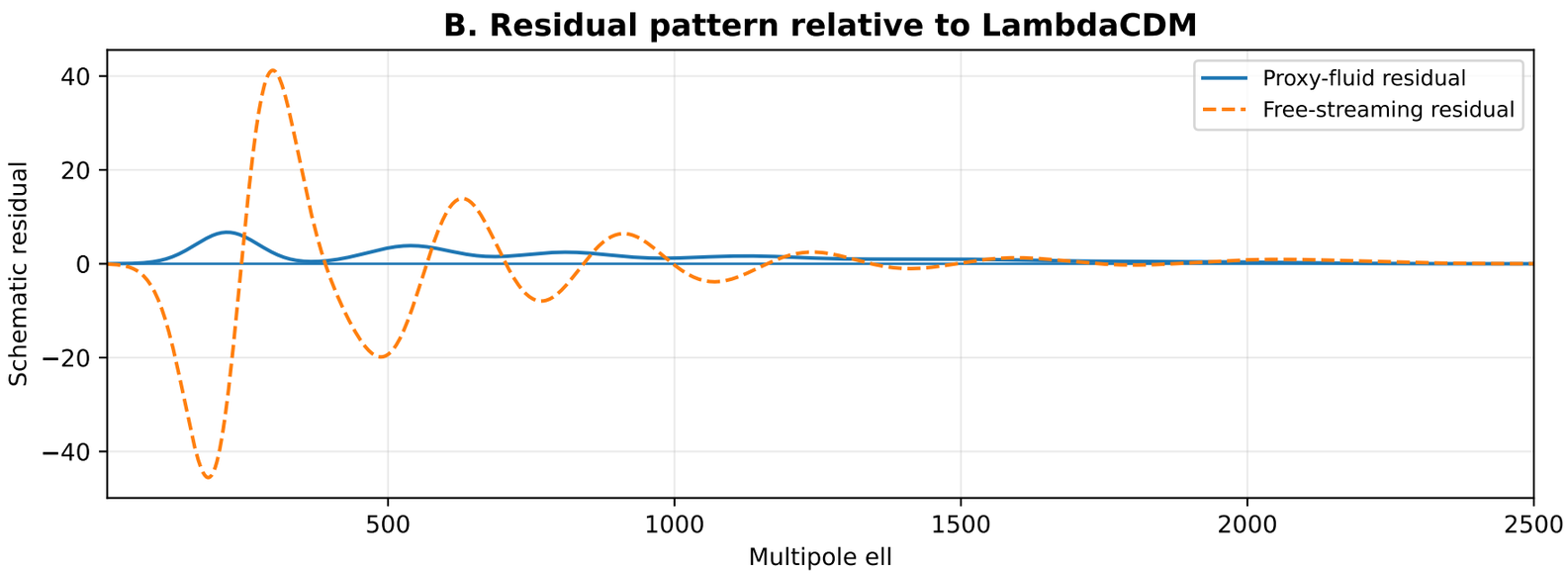
PTUM-F4: CMB Acoustic Phase-Shift Diagnostic

Extra radiation density is not enough: phase behavior distinguishes proxy-fluid and free-streaming radiation.







PTUM proxy fluid
w_psi = 1/3
c_s,psi^2 = 1/3
sigma_psi = 0
No neutrino-like phase shift

Free streaming
w = 1/3
c_s^2 = 1/3
sigma > 0
Neutrino-like phase shift

Figure PTUM-F4. A proxy-fluid relativistic component can add radiation density without producing the free-streaming acoustic phase shift. Curves are schematic diagnostics, not numerical CLASS output.